

Algebra II with Trigonometry

Chapter 11.4

Olympiad Solutions (GPT-5.4)

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Solutions

1. Find the center and classify the conic:

$$4x^2 + 9y^2 - 24x + 54y + 36 = 0.$$

Complete the squares:

$$4x^2 - 24x + 9y^2 + 54y + 36 = 0$$

$$4(x^2 - 6x) + 9(y^2 + 6y) + 36 = 0$$

$$4((x - 3)^2 - 9) + 9((y + 3)^2 - 9) + 36 = 0.$$

So

$$4(x - 3)^2 - 36 + 9(y + 3)^2 - 81 + 36 = 0,$$

hence

$$4(x - 3)^2 + 9(y + 3)^2 = 81.$$

Divide by 81:

$$\frac{(x - 3)^2}{81/4} + \frac{(y + 3)^2}{9} = 1.$$

Thus the conic is an **ellipse** with center

$$\boxed{(3, -3)}.$$

2. A parabola has focus $(5, -1)$ and directrix $x = -1$. Without writing the full equation first, find its vertex and the value of y when $x = 8$.

The directrix is vertical, so the axis is horizontal through the focus:

$$y = -1.$$

The vertex is midway between the focus $(5, -1)$ and the directrix $x = -1$. Since the distance from $x = 5$ to $x = -1$ is 6, the midpoint in the horizontal direction is

$$x = \frac{5 + (-1)}{2} = 2.$$

Hence the vertex is

$$\boxed{(2, -1)}.$$

The focal distance is

$$p = 5 - 2 = 3.$$

So the parabola has form

$$(y + 1)^2 = 4p(x - 2) = 12(x - 2).$$

Now set $x = 8$:

$$(y + 1)^2 = 12(6) = 72,$$

so

$$y + 1 = \pm 6\sqrt{2},$$

and therefore

$$\boxed{y = -1 \pm 6\sqrt{2}}.$$

3. An ellipse has center $(2, 1)$, passes through $(2, 5)$, and has foci $(2, -2)$ and $(2, 4)$. Find its equation.

The foci and center all have $x = 2$, so the major axis is vertical.

The center is $(2, 1)$, and the foci are $(2, -2)$ and $(2, 4)$. Thus

$$c = 3.$$

Since the ellipse passes through $(2, 5)$, and this point lies on the major axis, it must be a vertex. Its distance from the center is

$$a = |5 - 1| = 4.$$

For an ellipse,

$$c^2 = a^2 - b^2.$$

Thus

$$b^2 = a^2 - c^2 = 16 - 9 = 7.$$

With vertical major axis, the equation is

$$\frac{(x - 2)^2}{b^2} + \frac{(y - 1)^2}{a^2} = 1.$$

Therefore

$$\boxed{\frac{(x - 2)^2}{7} + \frac{(y - 1)^2}{16} = 1}.$$

4. A hyperbola has asymptotes

$$y - 3 = \pm 2(x + 1),$$

and one vertex at $(1, 3)$. Find its equation.

The asymptotes intersect at the center, so the center is

$$\boxed{(-1, 3)}.$$

Since the asymptotes are

$$y - 3 = \pm 2(x + 1),$$

the hyperbola is centered at $(-1, 3)$ and has either

$$\frac{(y - 3)^2}{a^2} - \frac{(x + 1)^2}{b^2} = 1 \quad \text{with} \quad \frac{a}{b} = 2,$$

or

$$\frac{(x + 1)^2}{a^2} - \frac{(y - 3)^2}{b^2} = 1 \quad \text{with} \quad \frac{b}{a} = 2.$$

Now the given vertex is $(1, 3)$, which lies horizontally 2 units from the center. So the transverse axis is horizontal, and

$$a = 2.$$

Hence the equation has form

$$\frac{(x + 1)^2}{a^2} - \frac{(y - 3)^2}{b^2} = 1.$$

Its asymptotes are

$$y - 3 = \pm \frac{b}{a}(x + 1).$$

Comparing with slope 2, we get

$$\frac{b}{a} = 2.$$

Since $a = 2$, it follows that

$$b = 4.$$

Therefore

$$\boxed{\frac{(x+1)^2}{4} - \frac{(y-3)^2}{16} = 1}.$$

5. Determine all real values of k for which

$$x^2 - 4x + y^2 + 2ky + 4k - 5 = 0$$

represents exactly one point.

Complete the squares:

$$x^2 - 4x = (x - 2)^2 - 4,$$

$$y^2 + 2ky = (y + k)^2 - k^2.$$

So the equation becomes

$$(x - 2)^2 - 4 + (y + k)^2 - k^2 + 4k - 5 = 0,$$

that is,

$$(x - 2)^2 + (y + k)^2 = k^2 - 4k + 9.$$

This is a circle centered at $(2, -k)$ with radius squared

$$r^2 = k^2 - 4k + 9 = (k - 2)^2 + 5.$$

For the graph to represent exactly one point, we need $r = 0$, i.e. $r^2 = 0$. But

$$(k - 2)^2 + 5 > 0$$

for all real k .

Hence there is **no** real k for which the equation represents exactly one point.

No real value of k

6. A conic is given by

$$9x^2 - 16y^2 - 54x - 64y + 19 = 0.$$

Find its center, identify the conic, and determine its asymptotes mentally by transforming it to standard form.

Group and complete squares:

$$9x^2 - 54x - 16y^2 - 64y + 19 = 0$$

$$9(x^2 - 6x) - 16(y^2 + 4y) + 19 = 0.$$

Now

$$x^2 - 6x = (x - 3)^2 - 9, \quad y^2 + 4y = (y + 2)^2 - 4.$$

So

$$9((x - 3)^2 - 9) - 16((y + 2)^2 - 4) + 19 = 0.$$

Thus

$$9(x - 3)^2 - 81 - 16(y + 2)^2 + 64 + 19 = 0,$$

hence

$$9(x - 3)^2 - 16(y + 2)^2 + 2 = 0,$$

or

$$16(y + 2)^2 - 9(x - 3)^2 = 2.$$

Divide by 2:

$$8(y + 2)^2 - \frac{9}{2}(x - 3)^2 = 1.$$

More usefully,

$$\frac{(y + 2)^2}{1/8} - \frac{(x - 3)^2}{2/9} = 1.$$

So the center is

$$(3, -2),$$

and the conic is a **hyperbola** with vertical transverse axis.

For asymptotes, from

$$16(y + 2)^2 - 9(x - 3)^2 = 2,$$

we ignore the constant on the right and set

$$16(y + 2)^2 - 9(x - 3)^2 = 0.$$

Then

$$4(y + 2) = \pm 3(x - 3),$$

so the asymptotes are

$$y + 2 = \pm \frac{3}{4}(x - 3).$$

So the final answers are:

- Center: $(3, -2)$
- Conic: hyperbola
- Asymptotes: $y + 2 = \pm \frac{3}{4}(x - 3).$